Guardians of the Future - Challenge Library Pilot Version 1.0

1. Community Heat Spot Mapping (v1.0)

Description

Cities heat unevenly. Concrete, asphalt, shade, vegetation, and building form create microclimates that affect comfort, health, and energy demand.

This challenge invites a Circle to map local heat patterns using simple thermometers or basic sensors. The goal is awareness and repeat observation, not scientific precision.

By measuring temperatures at selected locations and repeating those measurements over time, communities can track change, identify vulnerable spots, and strengthen local adaptation capacity.

Location Model (Pilot)

Each Circle defines X Anchor Points at the start of the challenge.

Each Anchor Point includes:

- Name
- Short description
- One-time GPS capture (for the pilot we might need something easier)
- Optional photo

Measurement sessions must be linked to one of these Anchor Points.

Spatial coverage (Measured Area) is calculated automatically in the system (for the pilot we do this manually) based on these locations.

Preset Validation Level

Level 1 – Trust-based (default). Optional Level 2 with uploaded logs or photos.

Preset Impact Metrics

- Measured area (km^2) – calculated automatically
- Number of measurement sessions
- Number of repeat cycles (same Anchor Points)
- Temperature delta between cycles (if applicable)
- Number of participants involved

Impact Translation Table

- Measured area (km^2) → Spatial coverage of sensing activity
- Measurement sessions → Active engagement signal
- Repeat cycles → Monitoring continuity
- Temperature delta → Local adaptation tracking
- Participants → Democratic Social Infrastructure signal

Validation & Reporting Flow

Circle marks “Heat Mapping Cycle Completed.”

Submission includes:

- Anchor Points used
- Number of sessions
- Optional reflection

2. Fixed Observation & Time-Lapse Documentation (v1.0)

Description

Climate action often unfolds slowly. Surfaces are removed. Trees grow. Compost matures. Insulation improves comfort. Biodiversity returns (or disappears). Yet change is difficult to perceive day by day.

This challenge invites a Circle to define a Fixed Observation Point — a precise camera position and viewing angle — and repeatedly document the same location over time. The collected images are assembled into a visual sequence that makes the transformation visible.

The goal is not measurement, but visibility.

Activation Models

A) Standalone Mode – The Circle selects a location and documents it over time.

B) Companion Mode – The Observation Point is attached to another active challenge (e.g., Schoolyard Naturalisation, Compost Sprint, Urban Greening, Household Heat Loss).

Observation Model (Pilot)

Each Circle defines one or more Fixed Observation Points, including:

- Location name
- One-time GPS capture
- Direction of view description
- Baseline reference photo
- Optional physical reference marker

Future photo submissions should be taken from the same position and angle.

No QR code or permanent infrastructure required.

Resources and Time Commitment

Tools:

- Smartphone camera
- Defined location and angle
- Repeat visits over time

Estimated time per photo: 1–2 minutes.

Recommended cadence: weekly, bi-weekly, or milestone-based.

Preset Validation Level

Standalone Mode: Level 1 – Trust-based.

Companion Mode: Inherits validation level of parent challenge.

Preset Impact Metrics

- Observation Points established (count)
- Photo submissions (count)
- Time span documented (days)
- Participants contributing (count)
- Time-lapse sequences generated (count, derived)

Impact Translation Table

- Observation Points → Structured place-based monitoring
- Photo submissions → Ongoing engagement signal
- Time span documented → Longitudinal visibility
- Participants → Democratic Social Infrastructure signal
- Time-lapse sequences → Narrative asset creation

Validation & Reporting Flow

Phase 1 – Registration

Circle defines Observation Point and uploads baseline image.

Phase 2 – Repeated Capture

Participants upload photos, system timestamps entries.

Phase 3 – Compilation

System generates slideshow or time-lapse sequence.

In Companion Mode, compilation may align with parent challenge milestones.

Impact Domains Activated

- Democratic Social Infrastructure
- Resilience (awareness and witnessing)
- Indirect support to Mitigation and Regeneration

3. Household Heat Loss Scan & Quick Fixes (v1.0)

Description

Many homes lose heat through avoidable weak points: poorly sealed windows and doors, thermal bridges, uninsulated attic hatches, and gaps around pipes or shutters. This increases energy demand, costs, and emissions.

This challenge enables households to see heat loss using a phone-attached IR device, identify major problem areas, and implement simple, low-cost fixes. The model follows:

Scan → Fix → Re-scan.

Impact is not only claimed — it is shown through before/after evidence.

Tools and Time Commitment

Tools:

- Phone-attached IR device (loaned by Guardians)
- Compatible smartphone
- Basic materials for quick fixes (weatherstripping, sealant, radiator foil, etc.)

Time:

Baseline scan: 30–60 minutes

- Quick fixes: 1–3 hours
- Re-scan: 15–30 minutes

Scan Point Model (Pilot)

Each household defines 6–10 standard Scan Points selected from a preset list, such as:

- Window frame
- Door threshold
- Exterior wall corner
- Attic hatch
- Radiator wall area
- Pipe or shutter box entry

Each Scan Point includes a label and baseline IR image.

Re-scans must use the same Scan Points.

Preset Validation Level

Baseline Scan: Level 2 (device-based evidence).

Fix Completion: Level 2 (photo or brief proof recommended).

Improvement Claims: Level 2 by default; Level 3 if exceeding predefined savings threshold.

Preset Impact Metrics

- Households scanned (count)
- Scan Points measured (count)
- Households improved (count)
- Fixes completed (count)
- Before/after evidence sets submitted (count)
- Estimated seasonal energy loss avoided (kWh/season – derived)
- Estimated CO2e avoided (kg CO2e – derived)
- Volunteer hours contributed (hours)

Impact Translation Table

Fix type and count → Estimated kWh avoided (seasonal) using predefined conservative rule-of-thumb factors.

Estimated kWh avoided → Estimated CO2e avoided via predefined grid emission factor.

Households improved → Adoption and resilience signal.

Volunteer hours → Democratic Social Infrastructure signal.

All kWh and CO2e values are proxy estimates based on centrally defined assumptions.

Validation & Reporting Flow

Phase 1 – Activation

Circle registers households and device assignment.

Phase 2 – Baseline Scan

IR images uploaded for Scan Points (Level 2 validation).

Phase 3 – Quick Fixes

Checklist of fixes submitted with optional photo proof.

Phase 4 – Re-scan

Before/after IR comparison submitted. Impact is calculated automatically after validation.

Impact Domains Activated

- Mitigation & Circularity
- Adaptation / Resilience
- Democratic Social Infrastructure

4. Neighbourhood Compost Sprint (v1.0)

Description

Organic waste is one of the largest components of urban household waste. When sent to landfill, it decomposes without oxygen and produces methane — a potent greenhouse gas. When composted locally, it becomes soil, moisture retention capacity, and fertility.

This challenge invites a Circle to organise a time-bound composting activation (e.g., 30 days) to collect organic waste from participating households and transform it into usable compost.

The purpose is to:

- Divert organic waste from landfill
- Reduce methane-related emissions
- Produce soil for local greening projects
- Build visible collective action

Activation Models (Pilot-Friendly)

1. Centralised Hub Model – A shared compost site where households bring organic waste.
2. Distributed Household Model – Households compost individually and report volume collectively.
3. Partner Model – Organic waste collected and delivered to an existing composting facility.

(The system focuses on measurement logic, not the composting method).

Logging Model (Ease-Focused)

Participants may log either:

- Litres of organic waste diverted (e.g., number of 10L buckets), or
- Kilograms of organic waste diverted (if weighing is possible).
- For volume entries, the system converts litres to kilograms automatically using a predefined density factor set centrally.

Preset Validation Level

Level 1 – Trust-based (default for pilot-scale quantities).

Level 2 – Evidence-supported (optional photo of compost site, scale, or facility receipt).

Level 3 is not required for the pilot phase.

Preset Impact Metrics

- Organic waste diverted (litres or kg)
- Estimated organic waste mass (kg – calculated automatically)
- Estimated CO2e avoided (kg CO2e – derived)
- Participating households (count)
- Volunteer hours contributed (hours)
- Compost produced (litres or kg – optional)

Impact Translation Table

Organic waste (litres) → Estimated mass (kg) via predefined density factor

Organic waste (kg) → Estimated CO2e avoided (kg CO2e) via predefined landfill avoidance factor

Organic waste (kg) → Estimated methane avoided (proxy) derived from same factor

Compost produced → Soil regeneration output (direct measure)

Volunteer hours → Democratic Social Infrastructure signal

Participating households → Behaviour change spread

Validation & Reporting Flow

Phase 1 – Activation

Circle registers Compost Sprint and defines duration, model type, and collection method.

Phase 2 – Collection

Participants log litres or kilograms of organic waste, household participation, and volunteer time.

Phase 3 – Completion

Circle marks “Compost Sprint Completed.”

Submission includes totals and optional photos.

The system automatically calculates estimated mass and emissions avoided.

Impact Domains Activated

- Mitigation & Circularity

- Regeneration (soil production)
- Democratic Social Infrastructure

5. Schoolyard Naturalisation

Description of the Action and Why it Matters

We all remember schoolyards: places where we spent countless hours between classes. Yet today, many urban schoolyards are dominated by concrete and asphalt, with little or no green surface.

While this simplifies maintenance, it does not serve students, teachers, the surrounding community, or the urban ecosystem. A schoolyard could become:

- A sanctuary for urban biodiversity
- A space where intense rainfall is absorbed instead of flooding streets
- A cool oasis instead of a heat-reflecting concrete surface
- A living garden where students experience nature directly
- A space supporting both active play and quiet rest

The Schoolyard Naturalisation challenge supports schools and communities in transforming sealed schoolyards into regenerative, multifunctional green spaces.

Required Resources and Process Phases

Phase 1 – Site identification & assessment

- Access to schoolyard blueprint
- Mapping of existing surfaces
- Sun path analysis
- Regulatory requirement review

Phase 2 – Participatory discovery

- Workshop with teachers
- Workshop with students
- Workshop with parents

Phase 3 – Proposal & approval

- Formal proposal preparation
- Architectural validation (if required)
- Approval submission

Phase 4 – Fundraising

- Internal contributions
- Community fundraising
- External grants

Phase 5 – Implementation

- Removal of sealed surfaces
- Soil preparation

- Planting and installation
- Rainwater absorption features

Preset Impact Metrics

- m^2 of sealed surface removed
- m^2 of green/permeable surface created
- Estimated annual rainwater absorption (litres)
- Number of trees planted
- Number of native species introduced
- Volunteer hours contributed
- Amount of community funds raised

Note:

- Biodiversity index is relative (baseline vs. post-intervention).
- Volunteer hours and funds raised contribute to Democratic Social Infrastructure signals.
- Translation coefficients are predefined centrally by Guardians.

Impact Translation Table

Sealed surface removed (m^2) → Estimated rainwater retention (litres/year)

Green surface created (m^2) → Heat mitigation potential

Trees planted (count) → Estimated CO_2 sequestration (kg CO_2e/year)

Native species introduced (count) → Biodiversity enrichment index (relative increase)

Volunteer hours (hours) → Democratic Social Infrastructure signal

Community funds raised (€) → Local resource mobilisation signal

Impact Domains Activated

- Mitigation & Circularity
- Resilience & Biodiversity
- Democratic Social Infrastructure

Validation & Reporting Flow by Phase

Phase 1 – Activation

Milestones: Site identified, baseline documented, workshops conducted.

Reporting: Circle marks “Activation Complete”.

Submission: Short description + optional photos.

Validation Level: Level 1 – Trust-based.

Phase 2 – Deepening

Milestones: Proposal prepared, approval secured, fundraising initiated.

Reporting: Circle marks “Approval Secured”.

Submission: Proposal summary + approval confirmation + fundraising amount.

Validation Level: Level 2 – Evidence-supported.

Phase 3 – Expansion

Milestones: Physical transformation implemented.

Reporting: Circle marks “Implementation Completed”.

Submission: Before/after photos, m² transformed, planting list, volunteer hours, funds spent.

Validation Level: Level 2 default; Level 3 if impact exceeds predefined threshold.

Phase 4 – Institutionalisation

Milestones: Maintenance plan adopted, curriculum integration.

Reporting: Circle marks “Institutionalised”.

Submission: Maintenance summary + curriculum note.

Validation Level: Level 2 – Evidence-supported.

Progress Patterns Alignment

Activation → Deepening → Expansion → Institutionalisation

Impact aggregation increases in proportion to the number of completed phases.

6. Urban Environmental Sensing – Device-Based (v1.0)

Description

Understanding environmental stress in cities requires consistent, geo-referenced data.

Through temporary assignment of calibrated Guardians sensing devices, Circles measure temperature, humidity, particulate matter, and noise levels.

Measurement becomes meaningful when repeated at the same locations over time, enabling tracking of change and evidence-based resilience building.

Location Model (Pilot)

Each Circle defines X Anchor Points before measurement begins.

Each Anchor Point includes:

- Name
- Description
- One-time GPS capture (we need to find the easiest way to have these data)
- Optional photo

All device measurement sessions must be linked to an Anchor Point.

The measured (mapped) area (km²) is calculated automatically based on the spatial distribution of these Anchor Points.

Device Assignment Model (Pilot)

Each device has a unique ID and is temporarily assigned to a Circle.

All measurement data during the assignment period is automatically attributed to that Circle.

Preset Validation Level

Level 2 – Evidence-supported (automatic via device data).

But that still needs a human/admin validation after the Circle reports that they finished the challenge.

Preset Impact Metrics

- Measured area (km^2) – calculated automatically
- Number of device measurement sessions
- Number of repeat cycles (same Anchor Points)
- Environmental delta between cycles (if applicable)
- Number of participants involved

Impact Translation Table

Measured area (km^2) → Spatial coverage of sensing activity (mapped area)

Measurement sessions → Evidence collection signal (data collected)

Repeat cycles → Longitudinal resilience tracking

Environmental delta → Adaptation / intervention signal

Participants → Democratic Social Infrastructure signal

Validation & Reporting Flow

Phase 1 – Device Assignment

Device ID mapped to Circle.

Phase 2 – Data Collection

Measurement sessions conducted at Anchor Points.

Data automatically ingested.

Phase 3 – Summary & Reflection

Circle marks “Sensing Cycle Completed.”

Short summary optional.

Impact aggregation automatic.